## Slide 1: Session opening

**Main point:** Decide whether an apparent result deserves a forward trial.

A language model can extract information, form a forecast and help propose an action. These are different achievements. This session asks whether a historical experiment respects what could actually have been known and whether an improvement survives costs and later conditions. We will make the information timeline and the policy explicit before treating any attractive return as evidence.

**Example to use** A model can correctly read a revised earnings report today while still being unsuitable for claiming it predicted the original announcement.

**Assumptions and limits** Historical results need source, label and model availability checks. Illustrative examples are not claims of live market alpha.

**Ask the room** What would convince you that a result could have been obtained at the time?

## Slide 2: Why this section: can the result survive deployment?

**Main point:** Separate a successful demonstration from a credible prediction experiment.

The previous session asked whether adaptation changed behavior on held-out tasks. Here we ask whether the complete workflow could have made a useful decision before its outcome was known. A correct extraction is not automatically a good forecast, and a good forecast is not automatically a profitable trade. Each stage needs its own observation. A historical replay is valuable only when its information and execution assumptions are defensible.

**Example to use** An agent extracts a correct date from an article updated after the event. The extraction is correct today, but the replay would be using later information.

**Assumptions and limits** We must define the target period, available evidence, policy and costs. No stationarity or causal benefit follows from a good sample score.

**Ask the room** Which achievement is being claimed: extraction, prediction or decision value?

## Slide 3: Roadmap: from information to a credible decision

**Main point:** Explain why the sequence of checks matters.

First define the target. Next reconstruct what was knowable. Then compare complete workflows on the same opportunities and diagnose which component caused the result. Finally state the claim that the evidence supports. This order prevents us choosing the definition of success after seeing a favorable graph. The last step may be a restricted forward trial rather than full deployment.

**Example to use** If a return chart is attractive, first ask when its input signal became available rather than immediately selecting a larger position.

**Assumptions and limits** The experiment must keep a record of research choices and availability times; missing metadata limits the claim.

**Ask the room** Which of these checks is missing from your current evaluation process?

## Slide 4: Five outcomes for this session

**Main point:** Translate objectives into concrete deliverables.

By the end, you should be able to write the prediction target, reconstruct a valid information set, specify a fair baseline, separate probabilities from decisions and locate a failure along the chain. These are documents and tests a colleague can inspect. They are more useful than a general statement that a model is accurate.

**Example to use** A complete proposal states: forecast a five-day event using information available at the decision time, compare with an existing policy and include all rejected or missing cases.

**Assumptions and limits** Targets and costs must be fixed before the final test. Examples are illustrations rather than measured financial results.

**Ask the room** Ask someone to restate one objective as an artifact they would produce.

## Slide 5: Separate observation, prediction and action

**Main point:** Keep facts, forecasts and policy choices distinct.

Observation records what the source says. Prediction estimates an outcome not yet known. A policy turns that estimate into an action under costs and limits. Measurement then records what happened after the action. An LLM may assist several stages, but combining them in one paragraph does not remove the distinctions. When results change, we need to know which stage changed.

**Example to use** An agent observes a supplier delay, predicts a delivery risk and recommends reducing an exposure. The delay is evidence; the future delivery outcome is a label; the exposure adjustment is a policy.

**Assumptions and limits** A prediction is not a causal estimate of how an intervention changes outcomes. That requires extra assumptions or evidence.

**Ask the room** Which stage can a deterministic calculator verify, and which still needs empirical validation?

## Slide 6: Define the prediction before choosing the model

**Main point:** Explain the prediction equation and its time boundary.

X at time t is the information used now. The parameters also carry a time index because a model may be retrained. The output predicts an outcome over a later interval, from t to t plus the horizon. Define the unit, event and horizon before choosing training labels. Otherwise two systems can appear comparable while predicting different things.

**Example to use** For the machine example, one row is a machine at 09:00 and its label is whether it fails in the next seven days. For a stock, a row could target a five-day return above an agreed benchmark.

**Assumptions and limits** The target must be measurable and labels must resolve after the forecast time. The equation does not assume stationarity; transferring learned relations requires evaluation under the intended regime.

**Ask the room** When is the label for today actually available?

## Slide 7: A point-in-time labelled training record

**Main point:** Show a usable input and outcome label without future leakage.

Read this record in two stages. At the decision time we store the permitted evidence and its version. Only later do we append the realized outcome and the time that label became available. The label must not be included in the prompt used to make the original prediction. When retraining, only rows whose labels have already resolved can be used. An outcome label is different from a reviewer preference between two answers.

**Example to use** Fictional stock ABC: Monday 09:00 input is an archived release and prices then available. Label: one if the next five-trading-day excess return is positive, otherwise zero. The label is appended after that window resolves.

**Assumptions and limits** The benchmark, price convention, horizon, availability times and model vintage must be specified. This is an illustrative record; no actual return or trained model result is asserted.

**Ask the room** Which fields must be hidden at forecast time?

## Slide 8: Changing regimes: what must remain useful?

**Main point:** Explain why a historical relationship may fail later.

Markets and operating conditions change. A relationship learned from one period may no longer help in another. We do not need to pretend every input is stationary to write a forecast equation, but we do need evidence that the learned conditional relationship remains useful in the intended test period. Evaluate across stressed and ordinary periods, monitor later errors and define when the model must be reconsidered. An LLM can suggest regime indicators or hypotheses, but those suggestions are not validated merely because they sound plausible.

**Example to use** A liquidity warning extracted from news may matter differently during a funding crisis than during a calm period. Test the effect rather than assume one fixed mapping.

**Assumptions and limits** Generalization requires some stability or an effective adaptation mechanism. Drift tests cannot prove all future regimes are covered.

**Ask the room** What change would make the old relationship untrustworthy?

## Slide 9: Turn language into a reproducible measurement

**Main point:** Make a language-derived feature a defined object.

Record the source text, prompt, model version and parser so another analyst can reproduce the measurement. Saying we used sentiment is too vague. Specify whether the output is a category, duration, probability or structured fact. Also define what happens if the model returns an invalid format or the source is missing. Excluding those cases after the fact can make performance look better.

**Example to use** A feature could be the number of days of delay explicitly stated in an archived report, with missing rather than zero when no duration is provided.

**Assumptions and limits** The transformation must use only eligible information. Reproducibility does not prove the feature predicts the chosen outcome.

**Ask the room** How would a colleague reproduce one feature value six months later?

## Slide 10: Chronological rows do not guarantee historical knowledge

**Main point:** Distinguish event time from information availability.

An event can happen before the information describing it is published. A database row can also be revised later while retaining an old date. The information set at time t must therefore contain the versions actually obtainable then, not everything that now has an earlier event date. The feature construction must respect that set. Sorting rows by a date column alone does not establish this boundary.

**Example to use** A Monday event is reported on Wednesday. A Monday forecast cannot use the Wednesday article merely because its subject occurred on Monday.

**Assumptions and limits** Availability records and version history must be accurate. The equation assumes the feature function receives only that allowed information; it does not itself authenticate timestamps.

**Ask the room** Which timestamp in your dataset indicates when the workflow could first obtain the value?

## Slide 11: A saved URL is not a saved historical document

**Main point:** Explain vintage leakage through an everyday example.

A URL identifies a location, not necessarily a fixed version of the content. If that page is revised, fetching it today may return facts unavailable at the historical decision time. Save the actual text or file used, a version or hash, and evidence of when it was available. A hash helps identify a saved version but does not by itself prove when it was published.

**Example to use** The slide shows demand growth revised from two percent to five percent at the same URL. A replay that fetches the revised page gives Monday knowledge of Tuesday updates.

**Assumptions and limits** The saved snapshot must itself be timely and provenance must be reliable. A newly saved copy of an old-looking page is insufficient.

**Ask the room** Could you reconstruct the exact paragraph seen by the original decision?

## Slide 12: The model has a knowledge history too

**Main point:** Explain leakage that document filtering cannot remove.

A model trained after an event may have absorbed facts about how it ended. Restricting retrieved documents does not remove that information from its weights. This makes a current model applied to old events a different experiment from a strategy that could have operated then. You can still study extraction retrospectively, but narrow the claim accordingly. A prospective trial with frozen settings is often more defensible for prediction.

**Example to use** A current model may recognize a company and remember its later collapse even when the prompt contains only earlier filings.

**Assumptions and limits** Model release date is not a complete training-data audit. A claimed cutoff is useful metadata, not proof that no contamination exists.

**Ask the room** What additional evidence would you need to call this a historically feasible trading experiment?

## Slide 13: Question: which claim does the evidence justify?

**Main point:** Check whether source timing and model timing are being conflated.

Give the room time to choose. The scenario filters documents for a past decision but uses a later model. The filter is a useful control over supplied context. It cannot establish what was already encoded in the model. Ask participants to identify the exact unsupported step in the strongest claim rather than dismiss every retrospective study.

**Example to use** A 2026 model can produce an informative analysis of 2018 documents without proving that the same capability was available in 2018.

**Assumptions and limits** The question concerns model-vintage uncertainty, not whether every later model necessarily memorized every outcome.

**Ask the room** Which answer describes the limitation without overstating it?

## Slide 14: Answer: B — the model vintage remains unresolved

**Main point:** State the strongest defensible claim.

B is the appropriate conclusion. The result can support a retrospective finding under the specified inputs and model. It does not prove historical foresight. Report the unresolved model-knowledge boundary, then design a forward trial or use a defensible model vintage if available. This is a limitation on the claim, not an assertion that the analysis has no value.

**Example to use** Keep the model and prompt fixed today and timestamp predictions for events that have not yet resolved. Those outcomes cannot already be in that frozen checkpoint.

**Assumptions and limits** Even a forward trial needs source integrity, valid labels and controls over any live tools that might expose future data during later replay.

**Ask the room** What would you freeze before beginning that forward trial?

## Slide 15: Evaluate as the forecast origin moves forward

**Main point:** Explain walk-forward evaluation as repeated historical decisions.

At each origin, train or update using only eligible past examples, freeze the prediction and evaluate the following period. Then move the origin forward. This resembles how the workflow would actually be maintained. A rolling window keeps a recent subset; an expanding window retains more history. Neither choice automatically solves regime change. Record which one you use and when retraining occurs.

**Example to use** Train through January, predict February, then update using data available by the next origin before predicting March. Do not tune February predictions after seeing February outcomes.

**Assumptions and limits** Training inputs and labels must both be available at the origin. Overlapping windows can make results dependent, so uncertainty estimates must respect that.

**Ask the room** Would daily retraining have labels available for a seven-day outcome target?

## Slide 16: A past row may still have an unresolved label

**Main point:** Explain why input age is not enough for training eligibility.

The equation includes a row only when both the input and its outcome label are available by the training time. A Monday input with a seven-day outcome cannot supply a complete target for a Friday training run. The calendar date of the input is old, but its future window is unfinished. Reporting delays can push availability later still. This applies to supervised model fitting, calibration and choosing thresholds.

**Example to use** A Friday retraining job must exclude a Monday seven-day failure label even if the eventual label is stored in the database today.

**Assumptions and limits** Availability functions must represent actual knowledge times. Partial outcomes require a separately specified censoring or survival model rather than silently treating unfinished cases as negatives.

**Ask the room** Which example on the slide is eligible and why?

## Slide 17: Implement the split; do not mistake it for an audit

**Main point:** Explain what a chronological splitter does and does not verify.

TimeSeriesSplit generates ordered training and test indices. Its gap excludes a number of rows near the boundary. This is useful scaffolding, but it cannot inspect document versions, label resolution or model training data. Choose a gap from outcome overlap and availability rather than a convenient number. With irregular observations, a row count is not a fixed amount of time.

**Example to use** A gap of five rows could mean five minutes in one dataset and five weeks in another. Neither automatically matches a seven-day label horizon.

**Assumptions and limits** The splitter assumes the ordering supplied is meaningful. Event-interval overlap and group dependence may require purging or a custom split.

**Ask the room** What would you check outside the library call?

## Slide 18: Fit every learned transformation inside the training window

**Main point:** Explain leakage through preprocessing and calibration.

Any transformation learned from data can leak future information. A scaler, feature selector, prompt choice, probability calibrator or decision threshold can all be fitted to later outcomes accidentally. Put those learning steps inside the appropriate development window and leave the final test untouched. If test failures are used to improve a prompt, that test has become development data.

**Example to use** Standardizing a feature using the full dataset lets future values influence the mean and variance used for earlier observations.

**Assumptions and limits** The pipeline must separate training, tuning, calibration and final evaluation according to the temporal task. Different roles can need separate eligible splits.

**Ask the room** Which part of your current preprocessing learns from all rows?

## Slide 19: A convincing comparison holds the opportunity set fixed

**Main point:** Explain why fair baselines matter.

Use the same opportunities, realized returns or events, fees and information boundary in compared conditions. Otherwise an apparent model benefit can come from easier cases or a different policy. Start with the existing operating approach and a simple baseline. Then isolate evidence quality, language interpretation and action rules using controlled comparisons. Keep abstentions and missing cases visible.

**Example to use** Compare two models on the same releases and trading dates, not one model on calm days and another on volatile days.

**Assumptions and limits** The comparison is meaningful within the chosen sample and assumptions. If intervention changes outcomes, replaying observed outcomes alone may not identify its causal benefit.

**Ask the room** What exactly is held fixed in your baseline comparison?

## Slide 20: Searching many designs changes what the best score means

**Main point:** Explain selection bias from research iteration.

If we try enough prompts, filters, models and thresholds, some will look good by chance. Selecting the highest historical return also selects favorable noise. Log the search, use development periods for choices and reserve a genuinely untouched evaluation. A confidence interval on the winning design alone may ignore how it was selected.

**Example to use** Trying one hundred thresholds and presenting only the winner is different evidence from fixing one threshold before seeing the results.

**Assumptions and limits** The research search process affects inference. Dependence among designs does not eliminate selection bias; correction or independent validation must match the actual search.

**Ask the room** How many alternatives were tried before this winning result appeared?

## Slide 21: Language likelihood and event probability answer different questions

**Main point:** Prevent token scores from becoming unvalidated forecasts.

A token score describes how likely a wording is under the language model and prompt. An event probability concerns whether a defined future outcome occurs. These can be related only through an explicitly evaluated mapping or predictive setup. A model can strongly prefer the word BUY without providing a reliable probability of a profitable trade. Candidate normalization does not change that distinction.

**Example to use** A seventy-percent score on the literal answer failure is not automatically a seventy-percent frequency of actual failures.

**Assumptions and limits** The event and horizon must be defined and outcomes independently labelled. Any calibrator must use separate eligible data and be checked after changes.

**Ask the room** What outcome would you count to validate the claimed probability?

## Slide 22: Evaluate probabilities and consequences separately

**Main point:** Explain the Brier score and why it is not profit.

For each binary event, subtract the observed zero-or-one outcome from the forecast probability, square the error and average across cases. Lower Brier score is better under this loss. It evaluates probability forecasts, including their calibration and resolution, but does not include every consequence of acting. A better score can coexist with a worse trading or inspection policy if thresholds, costs or severity differ.

**Example to use** A probability of 0.8 on an event that occurs gives squared error 0.04. A probability of 0.2 on the same event gives 0.64.

**Assumptions and limits** Probabilities must be in zero to one and refer to the same resolved binary target. Dependence affects uncertainty estimates; score comparison requires matched cases.

**Ask the room** Could a small probability improvement change no actual decisions?

## Slide 23: Choose a policy using explicit consequences

**Main point:** Derive the simple intervention threshold and its limits.

The general equation chooses an allowed action with the lowest expected loss under the estimated outcome distribution. The simple example compares inspecting now at fixed cost C with doing nothing and losing D if a failure occurs. If inspection completely prevents that loss, inspect when probability times D exceeds C. This is not a universal threshold; it follows from the declared cost and intervention model.

**Example to use** If inspection costs 100 and an unprevented failure costs 1000, the simplified threshold is ten percent. If inspection only partly prevents loss, that threshold changes.

**Assumptions and limits** The example is risk-neutral, uses one binary event, known fixed costs and perfect prevention, and omits capacity constraints and other consequences. Estimated probabilities and intervention effects must be validated.

**Ask the room** What assumption would fail first in your application?

## Slide 24: For trading, measure the return after implementation

**Main point:** Explain every term and omitted cost in the net-return equation.

The position selected at time t earns the next return. Changing the position incurs a linear cost proportional to the amount traded. This creates a simple connection from forecasts to an implementable strategy: frequent switching can consume a small predictive advantage. Keep exposure and turnover visible alongside the return. A signal can be informative yet not tradable at the assumed costs and latency.

**Example to use** A one-unit position earns ten basis points, but entering and exiting at six basis points each turns that isolated round trip negative before other costs.

**Assumptions and limits** The displayed model is a simplified single-asset, linear-cost accounting identity with ex ante positions. It omits or absorbs cash interest, funding, spread, slippage, market impact, capacity and execution delay as explicitly specified. Future returns never enter position selection.

**Ask the room** Which costs would make this example optimistic for your market?

## Slide 25: For risk, count severity and timing as well as errors

**Main point:** Explain why risk usefulness is more than an error count.

A warning must arrive early enough for an effective response and be useful after the cost of that response. Event frequency, loss severity and clustering matter. A model that misses one large event may be less useful than one that misses several small events. Repeated failures during a stressed regime can reveal a conditional weakness hidden by an overall average.

**Example to use** Two systems miss one event each, but one misses a minor delay and the other a major liquidity event. The identical error rate hides very different consequences.

**Assumptions and limits** Observed outcomes may depend on interventions. Comparing risk policies requires a credible model or study of what actions prevent or change, not just forecasts.

**Ask the room** Which consequence would be invisible in a plain accuracy score?

## Slide 26: Question: which improvement is decision-relevant?

**Main point:** Check whether participants ask for costs before selecting a policy.

Model A catches more failures but creates many more unnecessary inspections. Model B misses one additional failure but has fewer false alarms. Neither is automatically better without the severity of failures, the cost of inspection and the effectiveness of intervention. Ask the room to name the missing quantities rather than guess a preferred model.

**Example to use** A cheap harmless inspection can justify many false alarms. An expensive shutdown may not. The same confusion matrix can support different policies.

**Assumptions and limits** Both models must be assessed on comparable opportunities. The action model determines how predictive differences translate into value.

**Ask the room** Which extra number would you ask for first?

## Slide 27: Answer: C — specify the loss and intervention effect

**Main point:** Explain why no metric alone settles the decision.

C is correct because decision value depends on consequences. Count the outcomes after each action, including no action, and include delayed responses, false alarms and missed events. Report predictive metrics separately so users can see what comes from forecasting and what comes from the chosen policy. A policy can improve results by acting more conservatively without improving the forecast itself.

**Example to use** If each unnecessary inspection costs little while a missed failure is catastrophic, the higher-recall policy may be preferred. Reverse the costs and the choice may change.

**Assumptions and limits** The comparison needs declared costs and intervention effects. Do not infer causal prevention from a classification table alone.

**Ask the room** Ask someone to change one assumption and explain how the preferred policy might switch.

## Slide 28: Diagnose the chain before changing the model

**Main point:** Find which component needs repair.

Hold as much as possible fixed while changing one suspected failing component. Replace stale evidence, verify extraction against a reviewed label, recompute the policy from an unchanged signal and compare the intended order with execution. This separates the value of language interpretation from a gate or action rule. An evidence trace helps locate the relevant dependency; it does not automatically explain hidden neural computation.

**Example to use** If the model extracts the correct signal but the policy maps it to the wrong order, retraining extraction is not the first repair.

**Assumptions and limits** Interventions must be matched and expected behavior independently specified. Multiple simultaneous changes cannot cleanly identify one cause.

**Ask the room** Which single replacement would most directly test your current failure hypothesis?

## Slide 29: Class Exercise: did the signal predict or read the future?

**Main point:** Teach the information boundary through the actual controls.

Open the Alpha and risk exercise. Begin with the available-at-the-time signal and inspect one ledger row. Then choose the future-derived signal, first without and then with the availability gate. Observe which rows become ineligible. Finally change fees and compare gross and net results against no trade. The purpose is to distinguish a predictive achievement from a successful control against late evidence.

**Example to use** An apparently attractive signal can disappear when its timestamp shows it was produced after the decision. Blocking it demonstrates enforcement, not predictive alpha.

**Assumptions and limits** The replay is constructed for teaching. It does not establish market profitability, market impact or robustness of a real signal. The same opportunity ledger must remain fixed across conditions.

**Ask the room** Ask participants to show one blocked row and explain its timing in ordinary language.

## Slide 30: What to inspect before accepting the attractive result

**Main point:** Turn the exercise into a defensible conclusion.

Inspect when the signal was available, why any order was blocked, which costs were deducted and whether the opportunity set stayed the same. These fields tell us what changed. A gate can improve a simulation by preventing an invalid action without making the underlying model better at predicting. Keep the claim narrow enough to match the observation.

**Example to use** If a late signal is blocked, say the declared availability rule was enforced on that case, not that the system learned to predict the market.

**Assumptions and limits** Recorded timestamps and gating logic must be correct. Synthetic returns are designed outcomes, not independent evidence of real-market alpha.

**Ask the room** Which conclusion is justified by the row you just inspected?

## Slide 31: From a promising replay to a forward trial

**Main point:** Explain what to freeze before later outcomes become known.

Choose and version the target, sources, model, prompt, policy and costs before the trial. Timestamp predictions and include abstentions, rejected actions and missing data. Otherwise successful cases can be selectively retained. Define review triggers so a change in behavior or source quality leads to diagnosis. A forward trial is a learning step with limits, not an automatic deployment approval.

**Example to use** Freeze a weekly research workflow today and record every eligible opportunity for the next month before outcomes resolve.

**Assumptions and limits** Future data must not influence settings mid-evaluation unless the adaptation rule was specified and accounted for. Drift can invalidate earlier performance.

**Ask the room** What event would pause your trial rather than trigger an unreviewed retraining run?

## Slide 32: Return to the five learning objectives

**Main point:** Check whether participants can specify a credible experiment.

Ask for concrete answers: the target and horizon, evidence versions and availability, baseline, decision costs and a failure-isolation test. These pieces define what has been learned. A polished report without them may be impossible to interpret or reproduce. The session connects language models to the same disciplined prediction and decision questions required of any financial model, while adding special attention to language context and pretrained knowledge.

**Example to use** A complete evaluation card can fit on one page, but it should let another analyst reconstruct one decision and its label timeline.

**Assumptions and limits** Completeness of the card does not itself prove validity; the records and tests behind it must be checked.

**Ask the room** Which of the five deliverables can you now produce for your own workflow?

## Slide 33: Reading map: use each source for a specific decision

**Main point:** Explain what each external resource helps verify.

Use the reading map selectively. Forecasting references help design rolling evaluation and fair baselines. Real-time data documentation explains versioned historical information. Calibration material explains the difference between scores and event probabilities. Backtest-selection work addresses the effect of trying many designs. Read to resolve a specific modeling decision rather than treat a bibliography as evidence that the current experiment is valid.

**Example to use** If the concern is a revised economic series, start with version and availability documentation rather than a general transformer lecture.

**Assumptions and limits** The cited methods have their own scope and assumptions. Applying a library function or citing a paper does not audit the full workflow.

**Ask the room** Which source addresses the most important unresolved issue in your experiment?